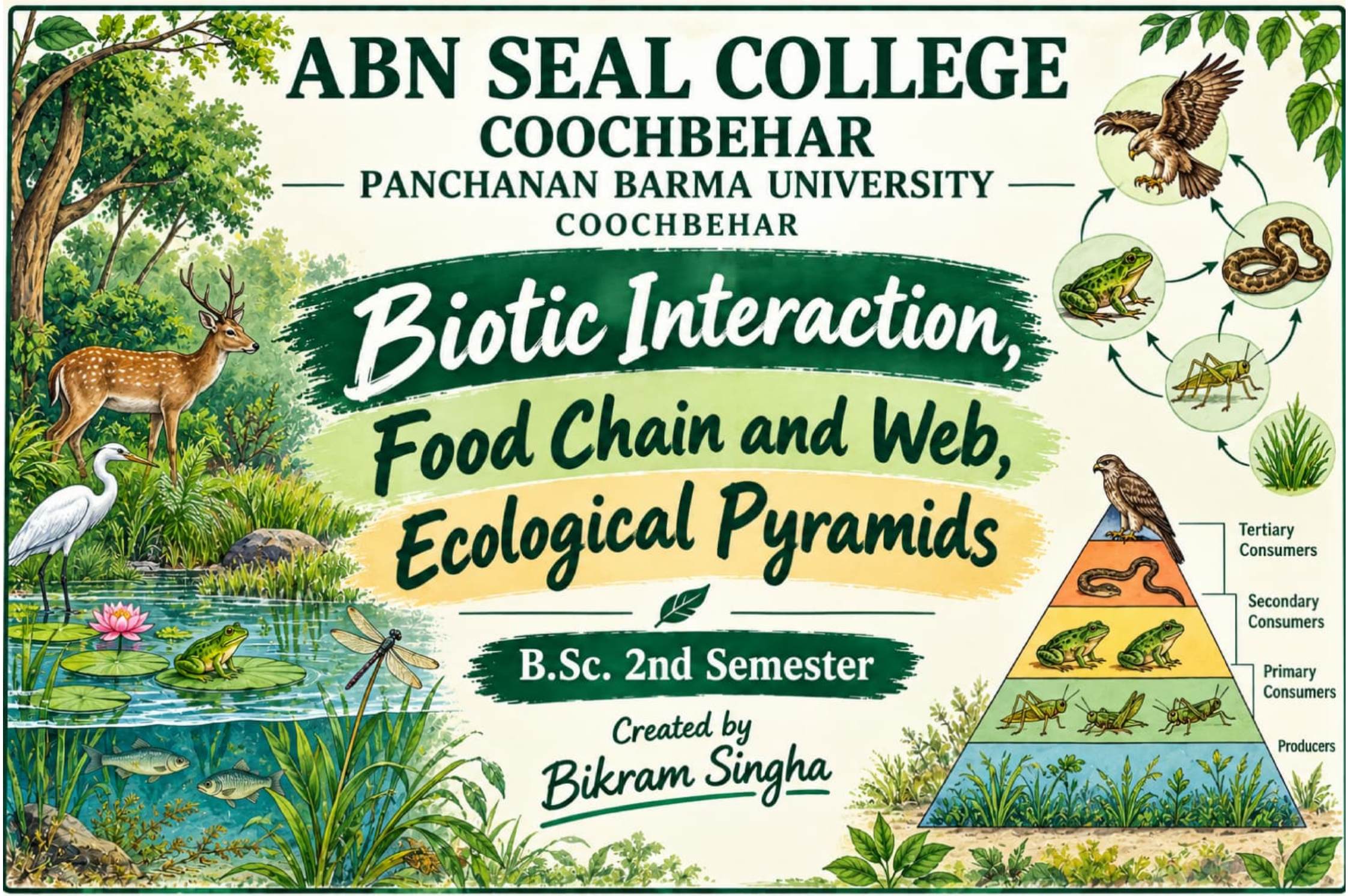
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Biotic Interaction, Food Chain and Web, Ecological Pyramids

B.Sc. 2nd Semester

Created by
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Ø Trophic organization-

Trophic organization refers to the arrangement of organisms according to their feeding relationships & the flow of energy from one organism to another.

The trophic organization in an ecosystem mainly devided into -

① Producer (Autotrophs) - Green plants & algae that prepare food by photosynthesis.

② Consumers (Heterotrophs) - Animal that depend on others for food.

- > primary consumers (Herbivores)
- > Secondary consumers (small carnivores)
- > Tertiary consumers (top carnivores)

③ Decomposer - Bacteria & fungi that breakdown dead organic matter & recycle nutrients.

example -

Grass → Grasshopper → Frog → Snake → Eagle

This sequence shows the transfer of energy through different feeding levels.

Ø Trophic Level-

A trophic level is each step or position in a food chain based on how organisms obtain their food.

Different trophic levels -

① First trophic level → producer
ex → Green plant, phytoplankton

② Second trophic level → Herbivores (primary consumers)
ex → Rabbit, deer.

③ Third trophic level → Small carnivores (secondary consumers)
ex → Frog, small fish

③ Fourth trophic level → Top carnivores (Tertiary consumers)
ex → Eagle tiger

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⑤ Decomposers → Act on all trophic level by decomposing dead organisms.
→ ex- bacteria, fungi.

Basis of Difference	Trophic organization	Trophic level
① Defination	The overall feeding arrangement in an ecosystem	A single feeding step in food chain.
② Meaning	shows the complete feeding relationship	shows the position of an organism.
③ Scope	Broad concept	part of trophic organization
④ Includes	producer, consumer, decomposer, food chain & webs.	producer lever, consumer level in food chain
⑤ Example	Grass → Deer → Tiger ecosystem relationship	Deer = second Trophic level.

Trophic organization → The whole system of feeding relationship
Trophic level → One step within that system.

ø symbiosis -

Symbiosis is a close long term association between two different species of organisms in which one or both organisms benifit.

example
- lichen - association b/w fungi & algae
- Rhizobium bacteria & leguminous plants.
- clownfish & sea anemone.

Types -

① ~~Mutuls~~ Mutualism → (+,+) both organism benifit from each other.

② protocopperation → (+,+) non obligatory ~~mutul~~ mutualism

③ Commonsalism → (+,0) one benifit, the other neither

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Symbiosis

Close, long term interaction b/w two or more biological species living together.

Key types

- Mutualism (+,+)
- protocooperation (+,+)
- Commonsalism (+,0)
- parasitism (+,-)

Other types

OTBO intimacy

Endosymbiosis
inside
> one organism lives inside the body of another
> bacteria inside gut

Ectosymbiosis
outside
> one organism lives on the surface of another
> lice on a host

OTBO survival

Obligate
> necessary for survival

Facultativ
> ~~not~~ Benificial, but not necessary for survival.

Population interaction / Biotic interaction

Positive ⊕

- mutualism (+,+)
- protocooperation (+,+)
- Commonsalism (+,0)

Negative ⊖

- Competition (-,-)
- predation (+,-)
- parasitism (+,-)
- Amensalism (-,0)

Neutral ⊙

- Neutralism (0,0)

Species A	Species B	
(+)	(+)	Mutualism
(+)	(+)	protocooperation
(+)	(0)	Commensalism
(-)	(0)	Amensalism
(+)	(-)	predation
(+)	(-)	parasitism
(0)	(0)	~~Mut~~ Neutralism
(-)	(-)	Competition

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Mutualism (+, +)

↓

Both species get benifit from each other

↓

Examples

① Rhizobium & Roots

↓

Heterotrophic N_2 fixing bacteria → provide nitrogenous product to plants

Roots → provide food to rhizobium

② BGA or Green Algae & Fungi

↓

BGA → provide food to fungi

Fungi → provide attachment to soil

Lichens

③ Fungi & Roots of higher plants

↓

Fungi → Helps in phosphorus absorption

Roots of higher plants → provide food to higher plants

Mycorrhiza

④ Fig & wasp

↓

wasp → provide safe place for egg laying

wasp → pollinate figs

⑤ Yucca & Moth

↓

Moth → provide safe place for egg laying

Moth → pollinate yucca

⑥ Mediterian orchid & Bumble Bee

↓

Bee → pollinate orchid

Orchid → provide pseudocopulati to complete its sexual desire (sexual decit

protocooperation (+, +)

↓

Non obligatory mutualism

↓

They will benifits from each other but can also survive without each other.

① Oxpecker & Rhino

↓

Oxpecker → provide relief from tics

Oxpecker → Also warn rhino from danger

Rhino → provide food to oxpecker

Plover & crocodile

↓

Plover → provide relief from blood sucking leeches

Crocodile → provide food to Plover

Sea Anemone & Hermit crab

↓

Sea Anemone → provide protection to crab

Hermit crab → provide movemen

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Commonsalism (+,0)

> One species got benifit but other species ~~nor b~~ neither ~~beniffi~~ benifitted nor harmed.

O ⟶ Orchid (+) ⟶ Mango tree (0)

B ⟶ Barnacle ⟶ whale

C ⟶ clown fish ⟶ Sea Anemone

C ⟶ cattle grazing ⟶ cattles
birds

Amensalism (−,0)

> one species is harmed but other species neither harmed nor benifitted.

① penicillin is ~~sect~~ secreted by penicillium, which kills other microorganisms but doesnot effect penicillium fungus.

② Streptomycin is secreted by streptomyces, kill other microorganism.

③ Roots of black ~~walt~~ walnut secrete chemical which kill other ~~microo~~ microorganism & plants.

Predation (+,−)

→ one species benifited & other is harmed.

→ Benifited species → predator

→ Harmed species → prey

① Predation is natures way of transferring energy from one trophic level to another trophic level.

② predator control or regulate prey population.

③ predator maintain species diversity by competition b/w species.

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Parasitism (+, -)

→ one species benifited & other is harmed
→ Benifited species → parasite
→ Harmed species → Host

eg → cascuta ~~plat~~ plant that twines around host plants & absorbs nutrients.

→ ticks on dog, sucks blood & nutrients from dog.

parasite → caseuta, ticks

Host → plants, Dog.

Impact of parasite on host

Parasite reduce the → Growth ↓, Reproduction ↓, Survival ↓ → Population density ↓ → more vulnerable to predatio

Types of parasitism

① Ectoparasite
↓
~~live~~ parasite live outside of the host body
ex- ticks, lice

② Endoparasite
↓
parasite live inside of the host body
ex- tape worm, round worm

③ Obligate parasite
↓
parasite must live in or on a host to survive
ex- many viruses & some protozoa

④ Facultative parasite
↓
parasite can live independently as a parasite
ex- some fungi & free living organism.

⑤ ~~Obligate~~ Temporary parasite
↓
parasites only visit the host briefly to feed
ex- mosquitoes feeding on blood

⑥ Permanent parasite
↓
parasite remain on or in the host for most of their life
ex- lice

⑦ Social parasitism
parasites exploit social behaviour of other species
ex- certain ant using another colony

⑧ Brood parasitis
one organism lays egg in another's nest, leaving the host to raise its young
eg- Cuckoo birds.

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Competition (-,-)

> Interaction of organism for same resources residing in same geographical area.

> Interaction in which fitness of one species is significantly lower in the presence of another species.

> It is generally believed that competition occurs when closely related species compete for the same resources but it is not entirely true.

→ Firstly -

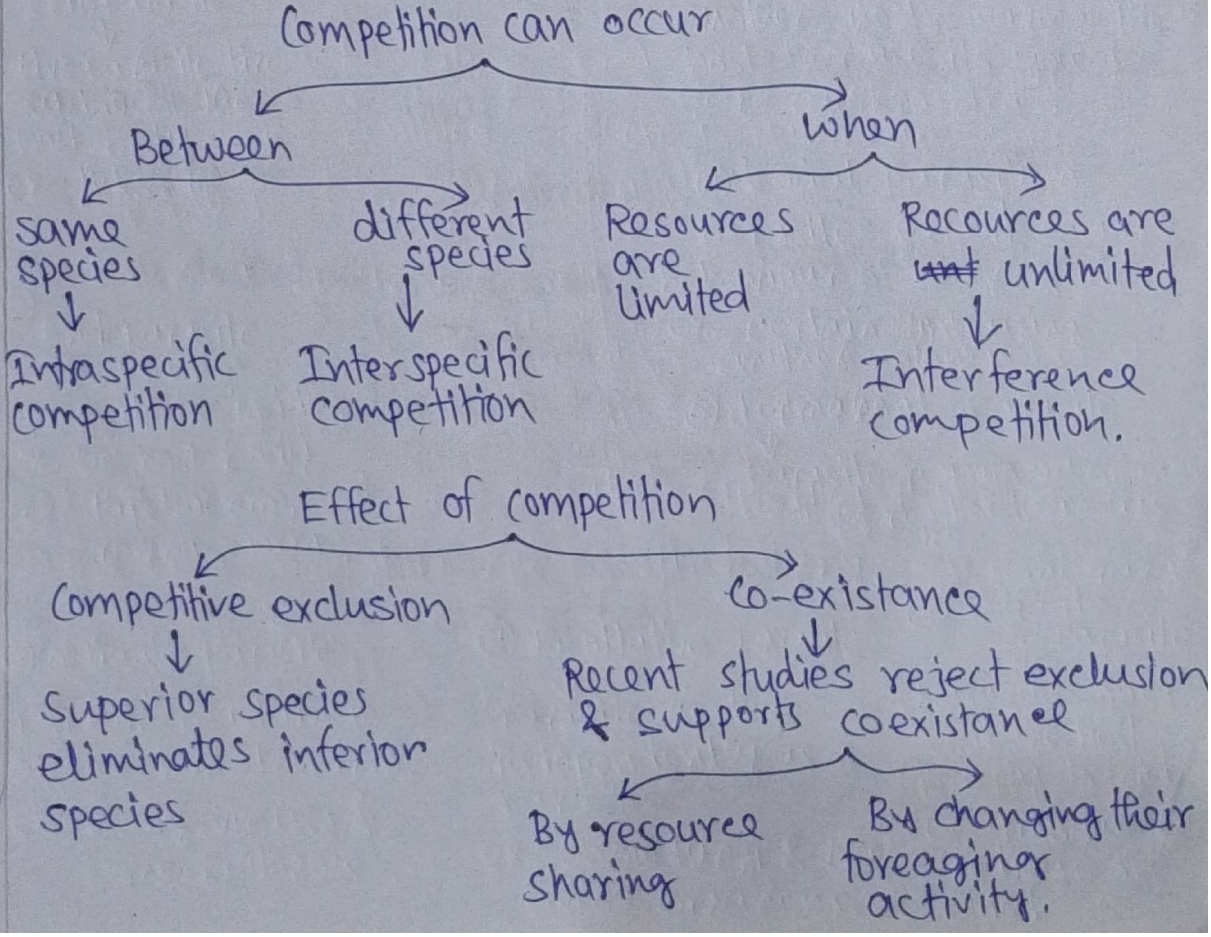
Totally unrelated species could also compete for the same resource

ex- In some shallow south American lakes visiting flamingo & residing fishes competes for same zooplanktons.

→ Secondly -

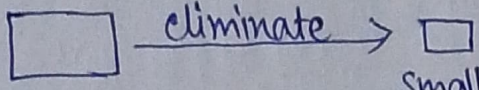
Recources need not to be limiting for competition to occur in interference competition.

↳ infinite resources.



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Competitive exclusion-

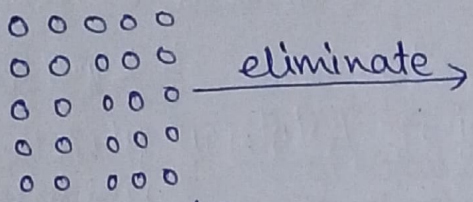


Big species

eg- Barnacle Balnus

small species

eg → Barnacle chathalmus



high number

eg- paramecium Aurelia

low number

eg- paramecium caudatum

Examples of Competitive exclusion

BGP

Barnacle Balnus → eliminates → Barnacle chathalmus

Goat → eliminates → Abingdon Tortoise

(Goat has high grazing effeciency)

↓

Tortoise extinct due to lack of food.

paramecium Aurelia. → eliminates → paramecium caudatum.

★ Last tortoise → Lonesome George → died in 2012

Ø

Gauss Competitive exclusion

↓

Two closely related species cannot survive indefinately

↓

Superior species eliminates inferion species

↓

B G P

eg of Gauss Competitive exclusion

> not the example of Gauss competitive exclusion

eg of Gauss Competitive exclusion

> But it is an example of competitive exclusion

R→ Because goat & tortoise are not related species.

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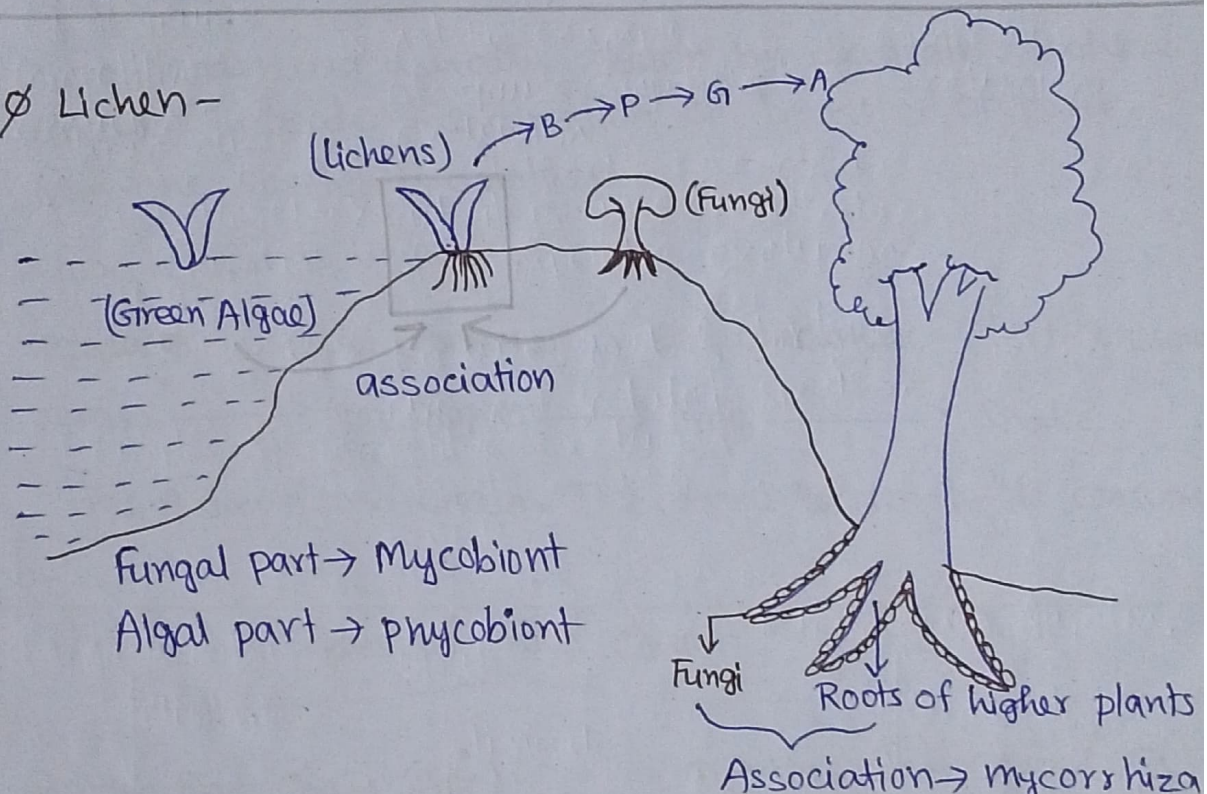
Ø Epiphytes -

Epiphyte is a type of plant that grows on another plant but does not take nutrients from it. It uses the host plant for physical support.

> They are not parasite
> Common in rainforests & humid environments.
> Often found growing on tree branches or trunks.

example -
orchids, ferns, mosses

Ø Lichen -



Basis of Difference	Intraspecific competition	Interspecific competition
1) Defination	Competition b/w individual of same species	Competition b/w individual of different species.
2) Type of organism involved	Same species	Different species
3) Requirement	Need for similar resource within the species	Need for similar resource b/w species.
4) Intensity	usually more intense	Usually less intense
5) example	Two deer competing for grass	One deer & one cow competing for grass.

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∅ Food chain-

> The route by which energy & nutrient transferred from the autotrophs through a series of organism that consume & are consumed.

> Plants are the main energy capturing unit of an ecosyste

> Energy stored by plants moves through the ecosystem in "a ~~series of sequence~~ series of steps of eating and being eaten". —Such sequence of series is referred as food chain.

> Food chain are usually shown by diagrams, where arrows indicate the direction of transfer of food, from one species to another.

> Food chain is linear.

producer pri. consumer sec. consumer. tert. consume

Grass ⟶ Grasshopper ⟶ Frog ⟶ Snake

Fig: A simple food chain. The arrow denote "is consume by"

Types of Food chain

Grazing Food Chain (GFC)

> Begins with green plants

> Sunlight is the source of energy.

eg→ Grass → Grasshopper
↓
Snake ← Frog
↓
Eagle

Detritus Food chain (DFC)

> Begins with dead plants & Animals.

> First breakdown by detrivores (eg- Bacteria, Fungi) then small animals than big animals.

eg-
Dead organic matter → Pri Detrivore (Bac, fungi)
↓
Carnivores (Birds) ← Sec. Detrivore (earthworm, snails)

Sun → producer → Herbivores → Carnivores

Grazing food chain

Detritus (energy) → ~~Detri~~ Decomposer → Carnivores

Detritus food chain

Fig: Grazing & Detritus food chain of an ecosystem.

Food web

> Food web is a network of interconnected food chain in an ecosystem.

> It shows or represents how different organisms are linked through various feeding relationships.

> It provides a more realistic ~~vo~~ view of how energy flows in nature.

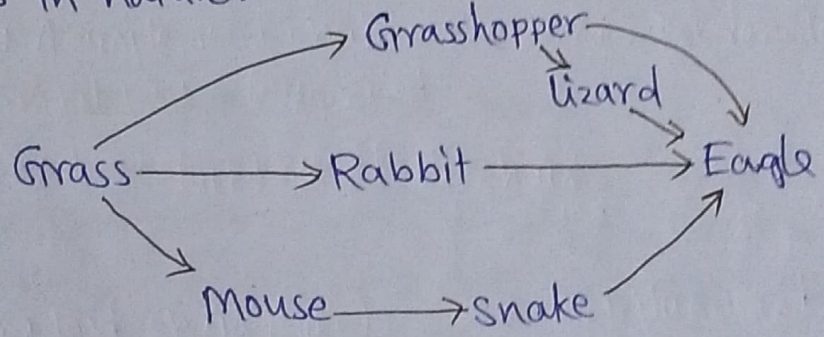


Fig: ~~Food~~ Food chain interlocked together making the ~~food~~ chain.

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Basis of Difference	Food chain	Food Web
① ~~Difference~~ Definition	A single linear pathway of food transfer	A network of interconnected food chain.
② Feeding relationship	simple & straight	complex & interconnec-ted.
③ Energy Flow	Follows one path	~~Foo~~ Follows many path
④ Stability	less stable	more stable
⑤ Number of organisms	Few organisms involved.	many organisms ~~involo~~ involved.
⑥ Effect of disturbance	one ~~dif~~ disturbance effects the whole chain	less affect due to alternative paths
⑦ example	Grass → Grasshopper ↓ Snake ← Frog	Multiple interconnected food chain in an ecosystem.

ø Role of decomposer in Food chain-

Decomposers are organisms that break down dead plants, dead animals, and ~~or~~ organic wastes into simpler substances,. Mainly bacteria & fungi act as decomposer

① Breakdown of Dead Matter-

They ~~doo~~ decompose dead bodies of plants & animals & convert complex organic matter into simple inorganic substances.

② Recycling of Nutrients-

Nutrients like Carbon, Nitrogen, phosphorus are returned to the soil & environment making them available again for producers (Plants).

③ Maintain Soil Fertility-

By adding Humus & nutrients to the soil, decomposer increase soil fertility an help plant growth.

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④ Complete the Food chain-

Decomposers form the final trophic level of the food chain by acting on dead remains of all organisms.

⑤ Prevent Accumulation of wate

They clean the environment by removing dead organisms and organic wastes

⑥ Maintain ecological balance-

Recycle of materials helps maintain balance in ecosystems & supports continuous energy flow.

ex- Bacteria, Fungi (mushrooms, molds)

Ø Biomass

Biomass is the total amount of living organic matter present in a particular area or at a particular trophic level at a given time. It includes all living plants, animals, and microorganisms.

In ecology, biomass generally refers to the dry weight of organisms because water content may vary from organism to organism. It is usually expressed as:

> grams per square meter (g/m^2)
> Kilograms per hectare (kg/ha)

Components of Biomass

① producers- plants, algae.
② Consumers- herbivores, carnivores.
③ Decomposers- bacteria, fungi.

Types of Biomass

Plant biomass	Animal biomass	Microbial biomass
> The total mass of plants in an ecosystem. eg- grasses in a grassland	> The total mass of Animals in an ecosystem. eg- fishes in a pond	> The mass of microorganisms such as bacteria and fungi

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Importance of Biomass

① Indicates the productivity of an ecosystem.
② Helps in studying food chains and ecological pyramids
③ Shows the amount of energy stored in living organisms.
④ useful in ecological & environmental studies.

∅ Standing Crop-

Standing crop is the total amount of living biomass present in an ecosystem or a particular trophic level at a given time.

It includes the mass of plants, animals & microorganisms present in that area.

Characteristics-
> Represents living matter only
> measured as biomass
> usually expressed as > g/m^2 or kg/ha

Example-
> Total mass of grasses in a grassland
> Total biomass of fishes in a pond.

∅ Standing State-

Standing state is the amount of inorganic nutrients present in an ecosystem at a particular time.

It includes nutrients like-
- Carbon
- Nitrogen
- Phosphorus
- Calcium
- water etc.

These nutrients are present in-
- Soil
- air
- Water

Characteristics
> Represents non living components
> measured in terms of nutrient concentrations.

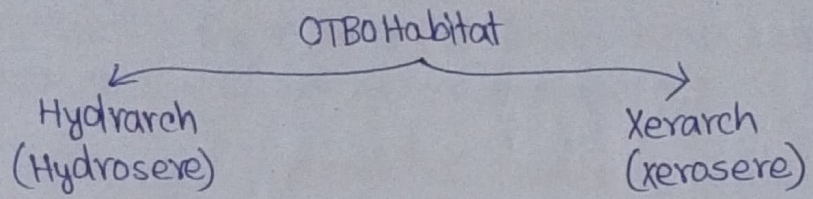
Example-
> Amount of nitrogen in soil.
> Quantity of CO_2 in atmosphere

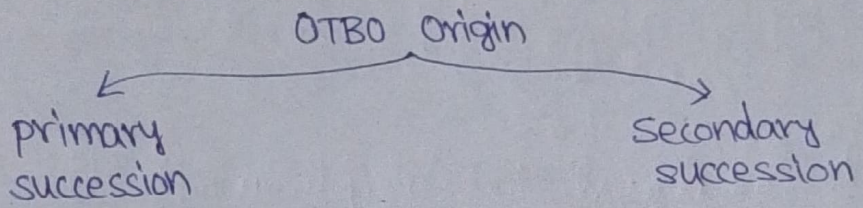
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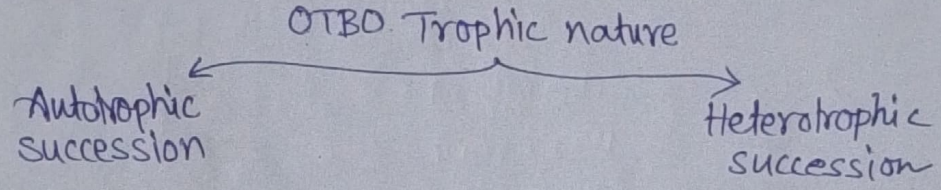
Basis	Standing Crop	Standing state
① Meaning	Amount of living biomass	Amount of inorganic nutrients.
② Nature	Biotic	Abiotic
③ Includes	plants, animals, microbes	minerals & nutrients
④ measure-ment	Biomass (g/m^2, kg/ha)	Nutrient concentration
⑤ Example	Biomass of forest trees	Nitrogen present in soil.

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Ecological Succession OTBO Habitat, Origin & Trophic nature -







Ecological Pyramid

» Ecological pyramid is a graphical representation of the quantitive relationship between different trophic levels of an ecosystem in terms of organism, biomass & energy present at each level.

» This graphical representation shows the trophic structure & functions at successive trophic levels, i,e from producer to Herbivore to carnivore, where the producer form the base of the pyramid & the successive level makes the apex.

» The concept was introduced by Charles Elton in 1927 & therefore it is also known as the Eltonian pyramid.

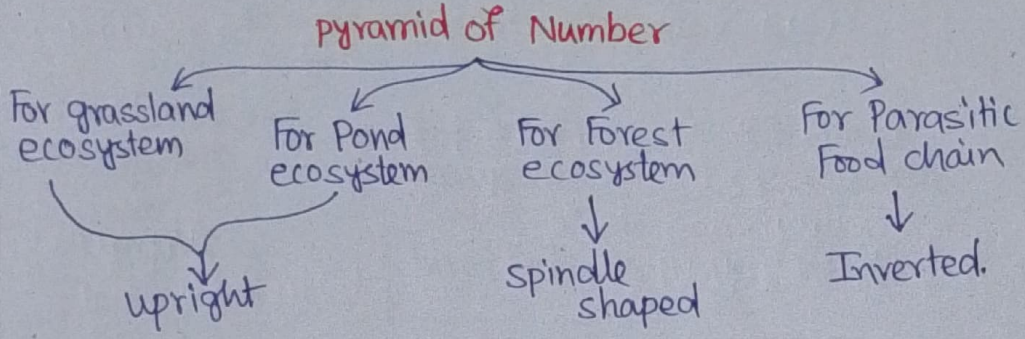
Types of Ecological pyramid

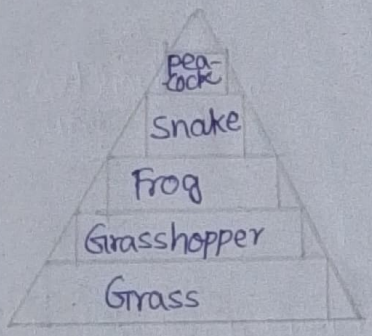
- Pyramid of Numbers → can be upright inverted or spindle shaped
- Pyramid of Biomass → can be upright or inverted.
- Pyramid of energy → Always upright

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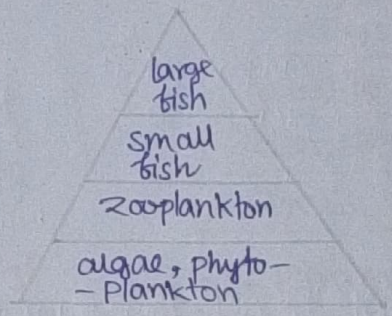
① Pyramid of Number-

A pyramid of Numbers is an ecological pyramid that represents the number of Individuals present at each trophic level of food chain or ecosystem.

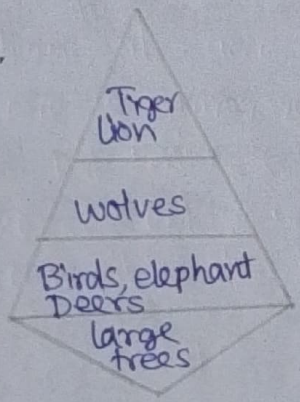




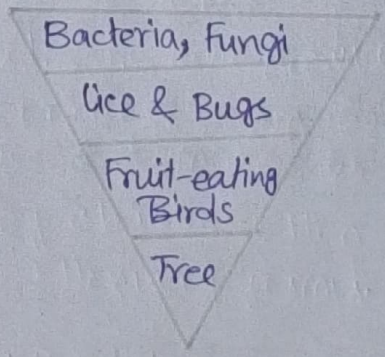
ⓐ Fig- Pyramid of Grassland ecosystem



ⓑ Fig- pyramid of pond ecosystem.



ⓒ Fig- pyramid of Forest ecosystem



ⓓ Fig- pyramid of parasitic Food chain

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② Pyramid of Biomass-

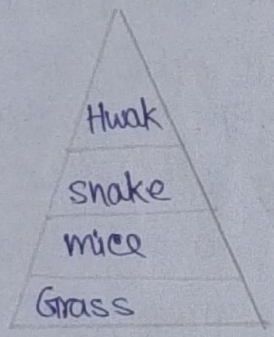
A pyramid of Biomass is an ecological pyramid that represents the biomass of an individuals present at each trophic ~~lee~~ level of food chain or ecosystem.

Pyramid of Biomass

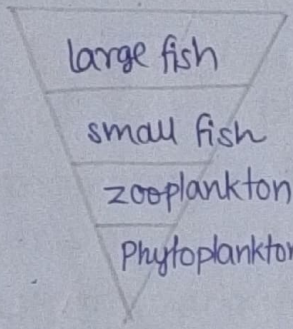
For Grassland ecosystem → upright

For pond ecosystem → inverted

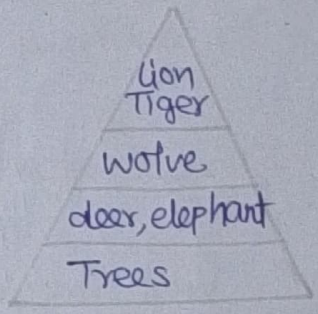
For Forest ecosystem → upright



ⓐ Fig- Pyramid of Grassland ecosystem



ⓑ Fig- Pyramid of pond ecosystem



ⓒ Fig- pyramid of forest ecosystem.

③ Pyramid of Energy-

pyramid of energy illustrates the flow of energy through different trophic levels in an ecosystem.

OF the three types of ~~ecos~~ ecological pyramids, the pyramid of energy give the best picture of overall nature of the ecosystem.

The shape of it is always upright, as in most of the cases, there is always a decrease in the energy levels at successive levels.

	no	Biomass	Energy
grasland	U	U	U
pond	U	I	U
Forest	I	U	U

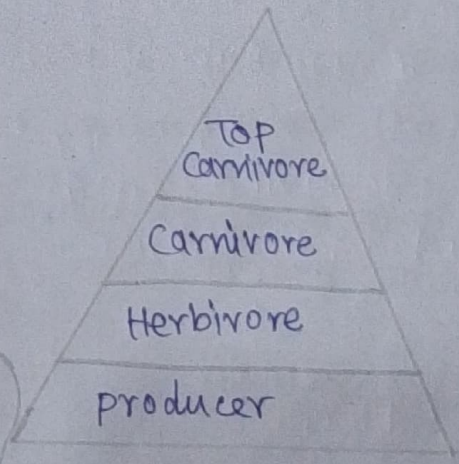


Fig- pyramid of energy

Pyramids of Number–

A pyramid of number is a graphical representation that shows the number of individuals present at each trophic level of a food chain in an ecosystem. It was proposed by charles Elton (1927).

① It represents the population size at different trophic levels.

② The base of the pyramid consists of producers while the upper levels consist of primary, secondary & tertiary consumers.

③ The width of each trophic level ∝ The number of organisms present at that level.

④ It helps in understanding the feeding relationships & distribution of organisms in an ecosystem.

⑤ The pyramid may be upright → as in grassland ecosystem where producers are most numerous

⑥ The pyramid may be inverted → as in parasitic food chain.

⑦ It doesnot consider the size or biomass of organism, it only indicates their number.

Pyramids of Biomass

① It shows the amount of living organic matter (dry mass) at each trophic level.

② Biomass is usually expressed in gram per square metre (g/m^2) or kilogram per square metre (kg/m^2)

③ The base of the pyramid is formed by producers followed by primary, secondary and tertiary consumer.

④ It provides a better indication of the quantity of living material than the pyramid of numbers.

⑤ For Forest & Grassland ecosystem → upright

⑥ For pond ecosystem → inverted.

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Q Why pyramid of energy is always upright -

A pyramid of energy is a graphical representation that shows the amount of energy available at each trophic level of a food chain in an ecosystem over a given period of time.

The pyramid of energy is always upright as-

① Unidirectional flow of energy-

In an ecosystem energy flow is unidirectional, that means one directional flow of energy. It flows from producers to consumers.

② Energy cannot be recycled within an ecosystem-

Energy doesnot return to the previous trophic level & cannot be recycled.

③ From First law of Thermodynamics-

First law of Thermodynamics stats that- energy cannot be created nor be destroyed, it can only be transfer from one form to another.

→Hence there was no chance of creating new energy in between the trophic level except producers.

④ From Second Law of Thermodynamics-

The second law of Thermodynamics stats that- during any energy transfer, a part of energy is lost in the form of heat, which increase the entropy of the system. Therefore no energy transfer is 100% efficient or 100% energy transfer is not possible.

⑤ Lindeman's 10% Rule-

The 10% Rule, proposed by Raymond Lindeman in 1942, it states that- Only 10% energy stored at one trophic level is transferred to the next higher trophic level while the remaining 90% is lost as heat & metabolic activity. activities.

All those parameter suggests that producer has higher energy and than primary than secondary & tertiary consumer.

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producer → pri consumer → Sec Consumer → Tertiary Consumer

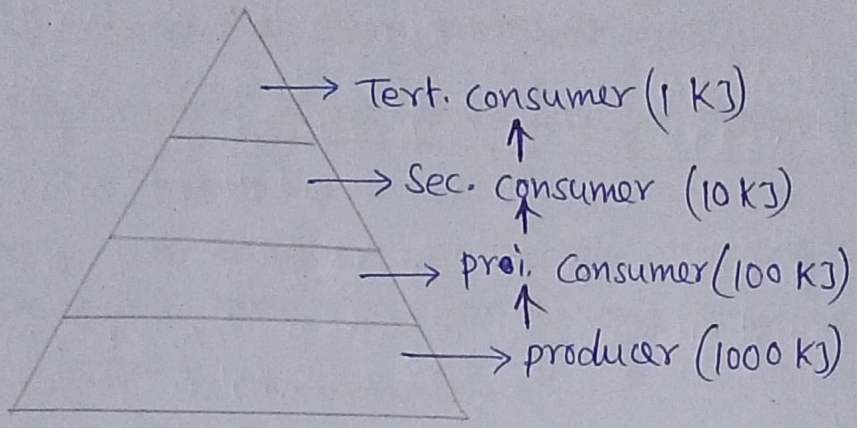
1000 KJ

$1000 \times \frac{10}{100} = 100$ KJ

$100 \times \frac{10}{100} = 10$ KJ

$10 \times \frac{10}{100} = 1$ KJ

(mathematical representation of 10% law)



[Fig :- Energy pyramid]

ø Upright v/s Inverted pyramid

Basis of difference	Upright pyramid	Invented pyramid
① Shape	Broad base & narrow appex	Narrow base & broad base.
② producers	Gratest in number/ biomass.	lessen in number/ biomass.
③ Stability	Generally more stable	Generally less stable
④ Energy flow	Follows normal ecological pattern	Doesnot represent energy flow.
⑤ Pyramid of energy	Always upright	never ~~inverve~~ invented.
~~⑥ Occurance~~	Commonly in ~~terrestrial~~ ecosystem	
⑥ Ecological interporetition	large producer base supports consumer	small producer base supports consumer by high reproduction
⑦ Example-	Biomass → Grassland, pond number → Grassland, Forest Energy → Always	Biomass → Parasitic FC Number → pond Energy → never.

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Lindeman's 10% Rule

In 1942, American ecologist, Raymond Lindeman proposed the 10% rule.

It states – only 10% of the energy stored at one trophic level is transfered to ~~he~~ the next trophic level, while remaining ~~9~~ 90% lost as heat, respiration, movement, excretion & other metabolic activities.

The 10% Law mainly come from Law's of thermodynamics

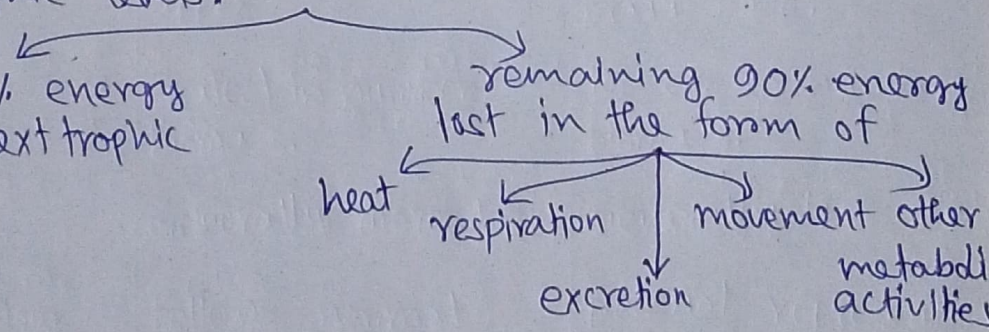
1st Law of Thermodynamics-

Energy can neither be created nor be destroyed it can only be transferred from one from to another.

2nd Law of Thermodynamics-

During every energy flow some energy lost in the from of heat, increasing the entropy of system.

↳ From those 2 ~~laaw~~ laws we can say that 100% energy stored in one trophic level cannot move to next trophic levels.



Example

producer produce (1000 Kcal energy)

↓

Herbivore got 10% of 1000 kcal

$= 1000 \times \frac{10}{100} =$ 100 Kcal energy

↓

Carnivore got 10% of 100 kcal

$= 100 \times \frac{10}{100} =$ 10 Kcal energy

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